

Construction & Cleanup (2)

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Question: Indicate the three main tasks/goals of a constructor. Provide a code fragment highlighting each of these tasks.

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Question: Describe the "Resource Acquisition is Initialization (RAII)" principle.

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Copy & Move (2)

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Question: Define the "Equivalence" and "Independence" criteria that must be met by copy operations.

Indicate a situation that may violate each of these two criteria.

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Question: Describe the function of the "delete" term when defining deleted functions. Indicate the main difference between the call of a deleted function and an undeclared function.

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RTTI (1)

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Question: Indicate what is the role of the "type_info" operator and why it is useful in the context of dynamic casting.

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Class Derivation and Inheritance (2)

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Question: Describe how the Member lookup process occurs in classes with single inheritance.

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Question: Given the class definitions in the code fragment below

- Indicate what specific operation or error occurs internally during the invocation of functions addNumber and addLocation.

- Justify the response given to each of these invocation lines.

```
class building{
public:
    double addNumber(int);
    void addLocation(string);
```

```
};

class house: public building{
public:
    double addNumber(double);
    void addLocation(char);
};
```

```
house h;
h.addNumber(111) ;
h.addLocation("Antwerp") ;
```

```
building b;
b.addNumber(7) ;
b.addLocation("Leuven") ;
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Object-Oriented Programming (1)
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Question: Indicate the main difference between "Subclassing" and "Subtyping"

- Provide an example illustrating this difference.

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Design Patterns (2)
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Question: Indicate the main purpose of the "Command" design pattern. Provide an example of an application/task where this is useful.

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Question: Describe the "Model View Controller (MVC)" design pattern.

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Generic Programming & Templates (2)
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Question: Describe the process of Deduction via Call Arguments during template argument deduction.

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Question: Indicate why is desirable to have the capability of "explicit specialization" for class templates.

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